■ Ed Tech Mobile App — Week 3 Deliverable

API / JSON Integration, Form Validations, & Asynchronous State UX

Program: Excelerate Mobile App Remote Internship (Cohort 0510)	**Team:** Team 6
Team Lead: Irtaza Ahmad (irtazaad328@gmail.com)	**Status:** Week 3 100% Completed
Repository: github.com/irtazaad328/ED-TECH_App	**Test Suite:** 27/27 Tests Passed (100%)

1. Screens Fetching Data from JSON / Mock API

During Week 3, static mock strings across catalog screens were completely converted into asynchronous, model-driven data architectures:

• Programs Listing Screen (programs_list_screen.dart):

- Asynchronously decodes assets/data/programs.json via rootBundle.loadString() mapped into immutable Program model instances.
- Utilizes Flutter's FutureBuilder<List<Program>> to manage load lifecycles without re-triggering network fetches unnecessarily.
- Supports real-time client-side search filtering across program titles, mentor names, and descriptions alongside category filter chips (Mobile Dev, AI, UI/UX, Cloud).

• Program Details Screen (program_details_screen.dart):

- Dynamically receives and unpacks the selected Program object via route arguments (ModalRoute.of(context)?.settings.arguments).
- Displays mentor profile, category pill tags, dynamic rating, student enrollments count, dynamic pricing, and an interactive 'What's Included' curriculum list.
- Features interactive bookmarking and link sharing actions with instantaneous user feedback.

• Cart & Enrollment Management (cart_management_screen.dart):

- Teacher-side administrative pipeline loading real-time student cart additions and paid enrollments.
- Provides segmented tab filtering (All, Paid, In Cart) and an interactive payment reminder dispatch action.

2. Forms Added & How They Work

Multiple interactive forms were architected utilizing Flutter's native Form and TextFormField widgets connected to isolated GlobalKey<FormState> controllers:

• Learner Login & Registration (login_screen.dart / signup_screen.dart):

- **Email Validation:** Enforces non-empty input and RFC-compliant regular expression pattern matching (^[a-zA-Z0-9._%+-]+@[a-zA-Z0-9.-]+\.[a-zA-Z]{2,}$).
- **Password Security:** Mandates a minimum 6-character length with non-empty checks and password obscurity toggle.
- **Confirmation Match & Terms:** Verifies password equality on registration and requires explicit Terms & Conditions acceptance.

• Teacher Authentication Forms (teacher_login_screen.dart / teacher_signup_screen.dart):

- Integrated with reusable AuthValidators utility providing immediate inline error prompts and SnackBar feedback.

• Multi-Method Checkout & Payment Form (payment_screen.dart):

- **Credit/Debit Card:** Sanitizes spaces and validates exact 16 numeric digits (^\d{16}$), MM/YY expiration format (^(0[1-9]|1[0-2])\/([0-9]{2})$), and 3-4 digit CVV/CVC.
- **Bank Transfer & Mobile Wallets:** Dedicated validators enforcing 11-digit national mobile format (^03\d{9}$).

3. Loading Indicators & Error Handling UX

• **Asynchronous Spinners:** Centered CircularProgressIndicator widgets styled in AppTheme.primary during initial asset and API parsing.

• **Self-Healing Error State:** When data parsing encounters issues, a user-friendly error card renders with an immediate **'Try Again'** retry button that re-triggers the loading future.

• **Double-Submission Prevention:** During form submission, an _isLoading boolean state disables all inputs and swaps primary button labels with a sleek white activity spinner.

• **Contextual SnackBar Feedback:** Floating SnackBars provide immediate alerts for invalid inputs, link copies, bookmark changes, and payment reminder dispatches.

4. Automated Widget & Unit Test Verification

Test Suite / File	Features Tested	Status
auth_validation_test.dart	Learner Login & Signup validation, email regex, confirm match, & spinner	**5/5 Passed**
teacher_auth_test.dart	Teacher Login & Signup validation, AuthValidators, & loading state	**5/5 Passed**
payment_validation_test.dart	Card 16-digit, MM/YY expiry, CVV, and mobile wallet number regex	**4/4 Passed**
cart_management_test.dart	Teacher cart tabs filtering (All/Paid/In Cart), search, & reminder action	**4/4 Passed**
learner_cart_test.dart	Dynamic subtotal, promo discount, item deletion, & empty state	**3/3 Passed**
program_details_test.dart	Dynamic route argument unpacking, bookmarks toggle, & share action	**4/4 Passed**
Overall Suite Total	**Full End-to-End Week 3 Verification Coverage**	**27/27 (100%)**